



Not a mass extinction

The impact that might have killed off the dinosaurs, could be the one that actually shaped the way for today's plant life on earth. <http://dgit.in/MassExtinction>

Chew to power up

A pair of researchers from Montreal have created a headset that harnesses the power of your jaws. <http://dgit.in/JawPower>

lites, to launch generational ships towards distant stars and create a new existence beyond Earth may seem to be just whimsical fancies – but its worth remembering so was the hope to feed India. It is only after it has already happened that we realise what is really possible.

The challenges facing sustainable human agriculture in non-Earth environments isn't however an intangible spiritual challenge. With significant problems and issues to be overcome in multiple fields of astrophysics, biology, genetic engineering, propulsion sciences, energy management and structural design we can only begin to scratch at the surface of what needs to be done. But it's a fun itch to scratch.

A Meal Fit For the Kings

Since the beginning of human space-flight and exploration food has been but another passenger on the journey. Precisely calculated rations for each crew member are a given cargo on every space flight with enough left over in case of emergencies. But with a launch cost of USD 13,000 (₹ 7,90,000) per kilogram on the Atlas V rocket it is perhaps fair to say that it is the most expensive meal eaten by any human. And even with the reduced launch costs from private space launch companies like SpaceX – at USD 4,000 (₹ 2,44,000) it's worth noting that we're still only floating in low-earth orbit at a maximum of 2,000 kilometers. Our ambition for the future is something a bit further.

Given these realities, the goal of implementing an inter-planetary human space mission can't possibly be contemplated with any seriousness until we have found a solution to the problem of feeding our passengers. In this light the idea of space farms and ecosystem vessels is highly appealing to both – our imagination and our accountants. The quest to build a sustainable environment within a space ship may seem ironic given how we treat our planetary ecosystem but it is the only viable option. Since plant based ecosystems are perfectly suited to create nutritious food, provide waste recycling, air and water purification, and further human psychological comfort – most research is dedicated to this goal.



Earth plant life is a product of complex inter-reactions over millions of years – adapting them for space is an uphill challenge.

All life, especially plant life, on Earth has evolved over millions of years under specialised factors such as solar energy, soil type, ecology, gravitation and other evolutionary laws. Only during the last few hundred thousand years of human civilization has an artificial process been developed to alter this natural trend. Techniques like selective breeding and genetic modification are the most recent of these experiments and even today, we are far from truly controlling all aspects of artificial agriculture. Under such circumstances, the problems associated with growing the simplest of vegetables can prove daunting.

Star Trek: The Search for Spinach

Plant life is affected by Earth's astrophysical attributes such as the gravitational forces and variable sunlight access. Any place in space would therefore offer radically different physical environments for which plant life would need to be adapted. The closest planet where we can consider setting up a "space farm" would be Mars, which receives just over half the solar energy as Earth does but is a vacuum environment. Cosmic greenhouses could compensate for the lack of atmosphere in its initial set up but the radically variable solar energy spread out across the planet would remain a challenge to any real plant growth. On the other hand, any location such as an orbiting station or Moon based greenhouse would receive significantly higher solar radiation than the Earth due to the absence of a pressurized atmosphere. As of now, no single crop could be adaptable to either or both of those extreme environments.

And referring back to the simple issue of gravity, we know that plants are dynamic organisms that have fluids flowing within them in one form or the other. Whether it's in their xylem or phloem, or within their cellular structure, the process has evolved to work in accordance with gravitational laws which would not be the same anywhere other than on Earth. In addition to this issue, the sheer vulnerability of non-Earth environments to other forms of cosmic radiation is also a perilous problem. Without the protective cover of an Earth like environ-



Mars is the closest and most likely option for humans to colonise and begin agricultural activities.